

Virtual Studies, Real Results? Assessing the Impact of Virtualization on Human-Robot Interaction

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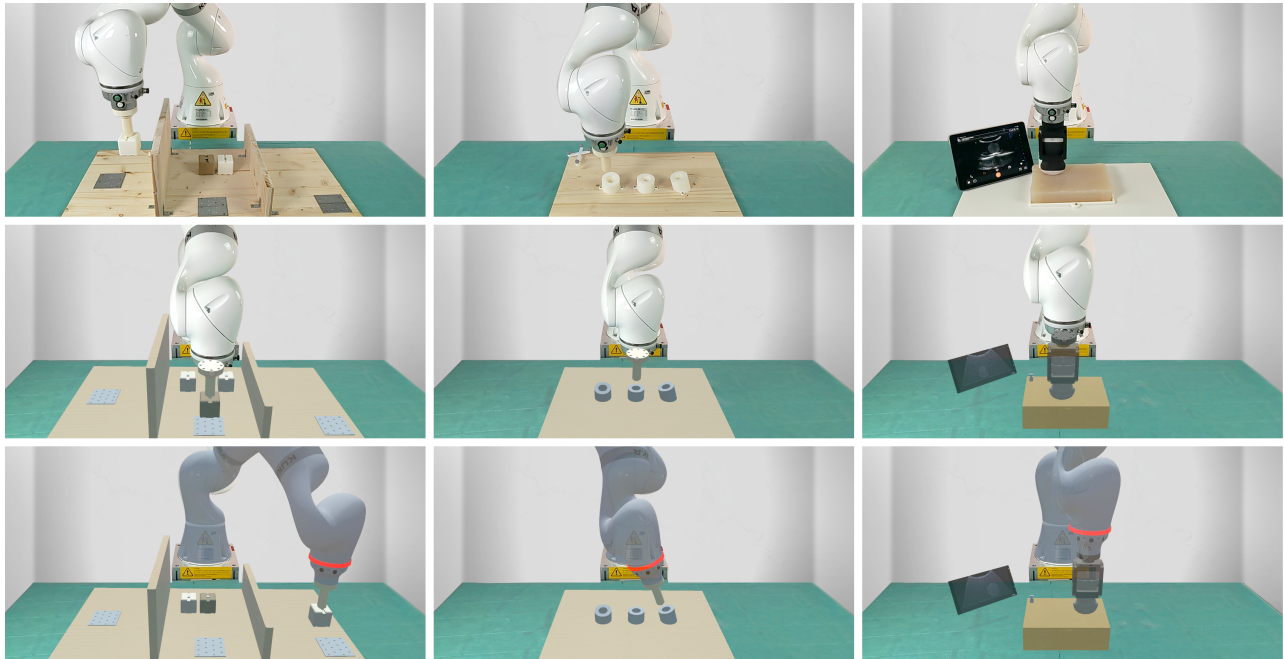


Figure 1: Overview of virtualization levels (rows: real, mixed, virtual) and tasks (columns: pick-and-place, peg-in-hole, robotic ultrasound)

Abstract

Extended Reality (XR) shows potential for human-centered evaluation of real-world scenarios and could improve efficiency and safety in robotic research. However, the validity of XR Human-Robot Interaction (HRI) studies remains underexplored. This paper investigates the transferability of HRI studies across virtualization

levels for three tasks. Our results indicate XR study validity is task-specific, with task virtualization as a key influencing factor. Partially virtualized settings with virtual tasks and a real robot, as well as fully virtualized setups with a simulated robot, yielded results comparable to real setups for pick-and-place and robotic ultrasound. However, for precision-dependent peg-in-hole, differences were observed between real and virtualized conditions regarding completion time, perceived workload, and ease. Demonstrating the task dependency of XR transferability and comparing virtualization levels, our work takes an important step in assessing XR study validity. Future work should isolate factors affecting transferability and assess relative validity in the absence of absolute validity.

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CCS Concepts

• **Human-centered computing** → **Empirical studies in visualization**; • **Computing methodologies** → **Mixed / augmented reality**; • **Computer systems organization** → **Robotic control**.

Keywords

Extended reality, Validation Study, Human-Robot Interaction

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1 Introduction

Extended Reality (XR) can serve as a valuable tool in robotic research for enabling education and training [32], teleoperation [24, 41], and the design of user interactions or robot appearances [39]. For the latter, virtualization serves as a tool for prototyping and design, helping to reduce experimental costs [23, 26], minimize danger during interaction [5, 25], and accelerate research and development [7]. Virtualizing robots also allows testing various setups without the need for expensive, robot-specific environments [39]. Additionally, using a virtual robot and task allows for the simulation of factors such as object recognition and task environments, while providing a robust and consistent setting [44]. Instead of virtualizing the entire setup, another approach is hardware-in-the-loop simulation, where a real robot is used in a virtual environment to enable fast prototyping or increase safety by reducing collisions [5].

In other areas, such as public displays [21], authentication systems [22], and assembly tasks [29, 38], virtual replicas of real environments have led to similar behavior and objective measures, highlighting the potential of XR as a testbed for human-centered evaluation of real-world scenarios. However, other studies show different perceptions across varying levels of virtualization [37]. While previous research has examined the similarity of perception between virtual and real robot scenarios [6, 8, 40] or introduced new interaction concepts in XR [24, 41], limited attention has been given to assessing the validity of XR user studies in Human-Robot Interaction (HRI), particularly for direct robot control. Efforts have been made to compare fully physical setups [27, 28] and physical robots in simulated tasks [5] with virtual representations of specific tasks, but a gap remains in evaluating the task-specific transferability across different virtualization levels. Addressing this gap is crucial, as perceptual differences between real and simulated robots [6, 40] could impact tasks differently, potentially mitigated by keeping the real robot in the loop while virtualizing the task. Therefore, this study examines the transferability of virtual robots, virtual tasks with real robots, and fully physical setups across three practical tasks.

2 Related Work

Plümer and Tatzgern [27] identified two approaches for validating XR in quantitative performance evaluations: a bottom-up approach,

isolating factors influencing performance [4, 14, 15], and a top-down approach, recreating real-world studies in XR for comparison [20, 22, 29, 37, 38]. In robotic research, the top-down approach has been the more prevalent method. Early works explored video representations of robots in user studies, where more positive interactions were found with physically present robots compared to virtual representations [3, 12, 17]. Additionally, participants tended to have a higher level of engagement with real robots compared to simulated ones [36]. However, Liu et al. [19] suggested potential advantages for XR, as they found superior task performance in XR compared to on-screen simulations.

In studies exploring XR in robotics, one focus is the perception of robots. For example, Robinette et al. [31] examined the perception of virtually, remotely, or physically present robots, finding comparable perceptions across different presence levels. Similarly, Han et al. [8] evaluated the comprehensibility of deictic gestures by a robotic arm, varying the virtualization of both the robot and the reference, and found no significant differences between virtual, real, and real-virtual misalignment. These findings align with Sadka et al. [33], who observed similar interpretations of robotic gestures between virtual and physical robots. Another set of studies focuses on the perception of robot proxemics. For instance, Li et al. [18] found that participants preferred closer distances with a real robot, while a virtual robot was perceived as more discomforting during HRI proxemics studies. In contrast, Inoue et al. [11] compared participants' impressions and feelings toward physical and virtual mobile manipulators passing by them, finding similar emotional responses in both cases. Weistroffer et al. [40] found no significant differences in robot perception and co-presence between real and virtual robots based on questionnaires. However, physiological measures such as heart rate and skin conductance were higher with real robots. Hsieh and Lu [10] investigated a handover task between a human and both a real and virtual robot, finding no differences in task performance across the two conditions. Similarly, Coguru et al. [6] compared the effectiveness of different safety zone visualizations for a real and virtual robotic arm, revealing differing preferences depending on the condition. Wijnen et al. [42] replicated a secret-keeping study with a real robotic guide and XR, revealing differing results and challenges in using XR for certain HRI studies. While the aforementioned related work contributes to understanding the perception of robots in physical and XR scenarios, it does not address interactive scenarios where the robot is user-controlled. Consequently, it may not provide sufficient insight into how interaction with a robot is influenced by the virtualization of the robot and its environment.

To compare interactive HRI with physical and XR robots, Nenna et al. [24] compared robot control in real and XR environments during a pick-and-place task, finding shorter operation times and lower implicit workload in XR, especially for high-precision tasks. Similarly, Whitney et al. [41] found differences in performance depending on the task, showing that while physical robots allowed for faster task completion, virtual robots enabled more accurate results in complex robot positioning tasks. However, both studies used different control schemes in virtual and physical environments, limiting the transferability of interaction methods between XR and physical robots due to variations in both virtualization and interaction techniques.

Plümer and Tatzgern [27] compared real-world and XR simulations for a drone steering task. Their results highlight differences in depth perception and perceived risk when interacting with a real and an XR environment. In a follow-up study, they compared the performance of steering a mobile medical robot in a real-world scenario versus a corresponding simulation [28]. Their results showed that, even when controlling for confounding factors, subjective feedback indicated that participants' attitudes toward the real and simulated devices differed, with higher stress and mental demand reported when controlling the real device. To investigate not only the impact of using a physical or simulated robot but also the effect of virtualizing the environment, Chen et al. [5] compared an XR environment with a real robot and a virtual task to a fully virtual simulation of the task. Their results indicated that participants rated the XR simulation as more realistic and better for producing reliable results. Additionally, task completion times were shorter in the fully virtual simulation, suggesting that participants were more cautious when interacting with the real robot. While these studies explore the differences between using virtual and real robots, as well as the use of a real robot in an XR environment, a direct comparison across all three stages is still missing. Additionally, these works focus on a single task each, limiting insights into how the task itself influences the transferability of interaction methods. To address this gap, we aim to compare the impact of virtualizing both the task and the robot across multiple tasks.

3 Material and Methods

This work assesses the transferability of user study results on HRI across different degrees of virtualization. Therefore, three tasks were implemented, each evaluated under three degrees of virtualization: a real robot with a real task (*real*), a real robot with a virtual task (*mixed*), and a virtual robot with a virtual task (*virtual*). A touchless interaction technique using a Leap Motion Controller 2 (Ultraleap, USA), placed in front of the user, controlled the robot in all scenarios. A pinch gesture activated control, allowing the robot's movements to follow the user's hand while the gesture was held. Translation maintained a 1:1 ratio, while rotation was scaled down proportionally at a 2:1 ratio, based on a preliminary study with five participants exploring different scaling ratios. The control system was implemented in Unity (Unity Software Inc., USA) and ran on a desktop PC.

3.1 Tasks

To assess varying ranges of motion, precision, and complexity, we explored the two most common HRI task domains: pick-and-place and assembly [30] and additionally investigated robotic ultrasound as a higher-level cognitive task.

Pick-and-Place: In this task, a cube is picked up from a start location and placed in a target location. To simplify interaction, a magnetic setup was used instead of a gripper, with magnets on the flange and object for pickup and additional magnets underneath for release onto a magnetic surface. Pick-and-Place was chosen as a task to require broad robotic arm movements, further encouraged by obstacles in the workspace.

Peg-in-Hole: Here, a cylindrical peg attached to the robot must be inserted into different holes. To require both translational and

rotational movements, the holes are angled in different directions. This assembly task was chosen for its demand for precise alignment and rotational movement.

Ultrasound: In robotic ultrasound, an ultrasound probe is attached to the robot, enabling imaging through robotic control. The task involves imaging a phantom with spheres of varying sizes and assessing their dimensions. Robotic ultrasound was chosen as it is an emerging field requiring human-robot interaction to complement autonomous control [35, 45] and to contrast with the two other industrial tasks. Additionally, it requires users to focus on both the robot and the sonographic images, adding cognitive complexity in contrast to the other tasks mainly addressing motor skills.

3.2 Real Environment

For the real environment (see top row Figure 1), a KUKA LBR IIWA robotic arm (KUKA AG, Germany) was used, with control software developed in KUKA Sunrise.OS. Components for the tasks, such as the cubes for the pick-and-place task, the holes for the peg-in-hole task, and the corresponding mounts on the robot, were 3D printed. For the ultrasound task, a Clarius probe (Clarius Mobile Health, Canada) was used, and the phantom and spheres, made of agar-agar with varying concentrations, were custom-built to exhibit different imaging properties.

3.3 Virtual Environment

To minimize confounding factors caused by mediated views through video see-through HMDs [27], an optical see-through HMD, the Microsoft HoloLens 2 (Microsoft Corporation, USA), was employed. The virtual environment, including 3D models of the robot and tasks, was also developed in Unity. Robot simulation was realized with ROS (Open Source Robotics Foundation, USA) using the Unity Robotics Hub. The movement sound of the physical robot was replicated with a velocity-dependent humming sound played through a speaker positioned beneath the robot. The ultrasound images were generated with a texture-based approach using a 3D model of the phantom for image creation, similar to Allgaier et al. [2]. Therefore, 3D models identical to the real phantom setup were built, and sonographic properties like noise and echogenicity were adjusted to closely mimic the real counterparts. Registration between the virtualized and the real environment was achieved through paired-point registration of a pivot-calibrated image marker attached to the robot, using the Vuforia SDK (PTC Inc., USA).

4 Experimental Methods

4.1 Preliminary Study

To assess simulation fidelity and identify areas for improvement, a preliminary study with five participants was conducted. While participants generally found the virtualized and real versions to be similar, some specific necessary improvements were identified. Additional sounds, such as the clicking of magnets during the pick-and-place task and scraping sounds for robot movements on surfaces, were incorporated. Moreover, static lighting conditions were adjusted to better mimic real-world lighting.

4.2 Study Design

4.2.1 Variables. The study employs a within-subject design, with the degree of virtualization (*real*, *mixed*, *virtual*) as the independent variable. Commonly used objective and subjective measures for product performance assessment [27] were utilized as dependent variables. The first variable was task completion time (TCT), measured from the start of the user's interaction to the robot's final movement before task completion was confirmed. A foot pedal press confirmed task completion for all tasks. Task completion for the pick-and-place task involved placing three cubes onto designated metal plates. In the peg-in-hole task, participants had to insert a peg into three holes, with a sound indicating successful insertion. For the ultrasound task, completion was defined by confirming the order of three spheres by size (ranging from 25 to 50 mm) by pressing a corresponding pedal when each sphere was visible. To ensure consistent task completion across virtualization levels such as triggering the sound during peg-in-hole at the same insertion depth or confirming sphere visibility in the ultrasound task, the physical tasks were optically tracked using a FusionTrack 500 camera (Atracsys LLC, Switzerland). As a subjective measure, perceived workload was assessed using the raw NASA TLX questionnaire, while subjective task difficulty was measured with the Single Ease Question (SEQ). Participants were also asked to rate the similarity between virtualization stages for each task using a five-point Likert scale.

4.2.2 Procedure. Ethical guidance for the study was sought from our institution's ethics committee. At the beginning of the study, participants received a brief introduction on the purpose of the study and completed a written consent form. They were then introduced to the robot control through a short training session. The first task block followed, starting with an explanation of the task and its goal. The order of tasks was counterbalanced using a Latin square. In each task block, the three degrees of virtualization were presented in a counterbalanced order. Each combination of task and virtualization level was performed twice: once for training and once for evaluation. Training sessions were included to familiarize participants with the task. After each evaluation trial, participants completed the NASA TLX and SEQ. Upon completing each task block, they answered the similarity question and participated in a short semi-structured interview. After all three tasks, a final semi-structured interview was conducted to gather general feedback.

4.2.3 Sample Design. In total, 31 participants (13 female, 15 male, 3 diverse) took part in the user study, aged between 20 and 36 years ($M = 25.94$, $SD = 3.60$). Among them, 26 were right-handed, and 5 were left-handed. The group comprised 18 university students, 9 PhD students, 3 engineers, and 1 postdoctoral researcher. Participants reported an average familiarity with HRI of 2.77 ($SD = 1.23$) and XR of 3.10 ($SD = 1.35$) on a 5-point Likert item, where 5 indicates high familiarity. They self-reported an average technical affinity of 4.19 ($SD = 0.65$) and motor skills of 3.81 ($SD = 0.54$) on the same scale.

4.2.4 Statistical Analyses. Several statistical analyses were conducted using R (version 4.4.0) to assess respective null and alternative hypotheses of the virtualization levels for each dependent measure. First, Shapiro-Wilk and Levene's tests were conducted

to assess normality and homogeneity assumptions on TCT measures. These assumptions were violated. Hence, robust repeated measures ANOVAs for within-subject designs based on trimmed means were calculated (see [43]) for TCT data. In addition, robust repeated measures ANOVAs were also conducted on the TLX scale-data and the ordinal SEQ data. Post-hoc paired sample Yuen's tests based on trimmed means with Holm corrections for multiple testing were conducted as pairwise comparisons in case of significant ANOVA outcomes. Finally, Bayes factors were calculated using Bayesian ANOVAs for repeated measures to further interpret statistical results by estimating how well the null or alternative hypotheses could be supported [9]. The Bayes factor BF_{10} results were interpreted according to the thresholds defined by Lee and Wagenmakers [16], indicating the strength of evidence.

4.3 Simulation Fidelity

For identifying potential confounding factors, we follow the XRP validation framework introduced by Plümer and Tatzgern [27] to assess differences between the real and simulated versions.

4.3.1 Visual Fidelity: In the XR conditions, participants wore an optical see-through HMD, allowing them to directly perceive the real environment, eliminating the need for a virtual representation of the user's hand or environment. To create accurate representations of the task and robot for the virtualized conditions, publicly available mesh files for the robot¹ were used, along with 3D models of the printed components and custom-made CAD models for other task parts.

4.3.2 Haptic Fidelity: Since the same touchless control method was used for the robot in all conditions, and no direct contact with the environment was required, the haptic fidelity remained identical across all conditions.

4.3.3 Audio Fidelity: To replicate the sounds of the physical robot during movement, a velocity-dependent humming sound was generated for the virtual robot. In the preliminary study, additional sounds were identified for the interaction between the virtual flange and task. These included the magnet clicking when picking up and placing the cube, as well as scratching noises when the flange collided with surfaces.

4.3.4 Interaction Fidelity: All conditions used the same interaction technique, hardware, and set-up, ensuring consistent interaction fidelity across conditions.

4.3.5 Functional Fidelity: The physical and simulated robots were set to have the same maximum velocity and acceleration, as well as similar stiffness behavior. The magnets for the pick-and-place tasks were also tuned to match the real ones' reaching distance. The movement of the robot and its interaction with the environment were validated by simultaneously moving the virtual and real robot. For the ultrasound image, a rendering approach shown to produce realistic images [2] was used, with the visual properties of the imaged structures tuned to mimic the real components.

¹https://github.com/ros-industrial/kuka_experimental

4.3.6 Data Fidelity: Objective and subjective data were collected identically across all approaches, as the same automated data collection and questionnaires were used in all conditions. To ensure consistent measures of task completion across the virtualization stages, the components in the physical setup were optically tracked, allowing the precise relative position of the task to the robot to be known.

4.3.7 Simulation Overhead: During the real conditions, participants did not have to wear the HMD to allow for a realistic assessment of the comparability between the real and virtual conditions, as a study with the real robot would not involve wearing an HMD.

5 Results

5.1 Statistical Results

The following outlines the results of the statistical tests, with descriptive values and the results of pairwise comparisons displayed in Figure 3. For the pick-and-place task, no significant main effects were observed on the TCT ($F = 1.09$, $p = 0.346$, $BF_{10} = 0.51$), TLX ($F = 1.12$, $p = 0.332$, $BF_{10} = 1.41$), or SEQ ($F = 2.73$, $p = 0.079$, $BF_{10} = 1.56$). Regarding the peg-in-hole task, significant main effects were observed on TCT ($F = 25.36$, $p < 0.001$), TLX ($F = 25.93$, $p < 0.001$), and SEQ ($F = 20.94$, $p < 0.001$). For the ultrasound task, no significant main effects were found on TCT ($F = 1.81$, $p = 0.181$, $BF_{10} = 0.34$), TLX ($F = 0.23$, $p = 0.77$, $BF_{10} = 0.11$), or SEQ ($F = 0.04$, $p = 0.951$, $BF_{10} = 0.16$). As shown in Figure 2, the descriptive results for the similarity scores indicate task-dependent differences across virtualization levels. While the virtualization levels are generally perceived as similar for the pick-and-place and robotic ultrasound tasks, perceived differences emerge between the real and virtualized scenarios in the peg-in-hole task.

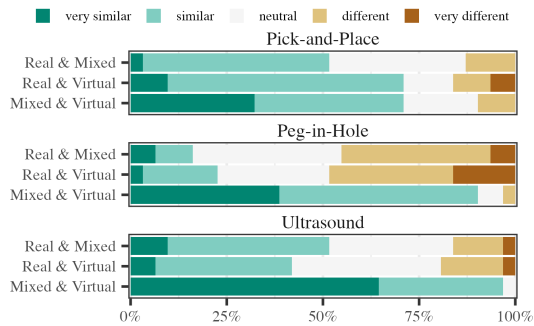


Figure 2: Plot of the similarity scores.

5.2 Contextualization of Results

No significant main effects were observed on TCT, TLX, or SEQ for the pick-and-place task. Similarity scores suggest that most participants perceived the task as similar across conditions and Bayes factor analysis provided anecdotal evidence supporting the null hypothesis for TCT, indicating no effect. However, while again no significant effects were found for TLX and SEQ, Bayes factor analysis (see BF_{10} in subsection 5.1) provided anecdotal evidence supporting the alternative hypothesis. This may be attributed to

differences reported during interviews, where 13 participants noted a greater fear of collision in the *real* scenario, and seven described being more cautious. Participants emphasized that virtualizing the task was the primary factor in reducing perceived risk.

For the peg-in-hole task, significant main effects were observed on TCT, TLX, and SEQ. Post-hoc comparisons showed significantly higher TCT and TLX scores, and lower SEQ scores, in the *mixed* and *virtual* compared to the *real* condition. These differences likely stem from perceptual challenges in the virtual tasks. Sixteen participants reported difficulties with the perception of depth and the peg's rotation, which made aligning the peg with the hole more challenging. Consequently, eleven participants described achieving precise motions as difficult. Although four participants described the perception of the real robot in the *mixed* condition as helpful, this did not result in significant differences between the *mixed* and *virtual* conditions which was also reflected in the high perceived similarity between the *mixed* and *virtual* conditions. This suggests that the use of a virtual task is the primary factor contributing to perceptual differences that result in increased task difficulty, longer completion times, increased workload, and decreased task ease.

The results for the ultrasound task show no significant main effects of the virtualization approach on all measures. Post-hoc Bayesian analysis indicates anecdotal evidence supporting the absence of an effect for TCT and moderate evidence for TLX and SEQ. Differences between virtualization approaches may stem from variations in the ultrasound images, with 11 participants noting the real image had more noise. Additionally, nine participants reported that pressure was harder to perceive in the *mixed* and *virtual* scenarios, with eight attributing this to less complex feedback, such as reduced deformation of the phantom. Conversely, nine participants noted being afraid of applying too much pressure in the *real* scenario. Similarity scores emphasize the resemblance between the *mixed* and *virtual* scenarios.

6 Discussion and Conclusion

The goal of this work was to assess the transferability of HRI study results between real and XR setups. Our results highlight that the XR validity depends on the task being investigated. For broader movement tasks, such as the pick-and-place task, and complex tasks such as robotic ultrasound, evidence suggests that conducting the task in a real versus partially virtualized setting yields comparable results. However, significant differences between the real and the XR setups were found for the peg-in-hole task, which requires higher precision. The primary factor behind this difference seems to be task virtualization, as using a real robot with a virtual task still resulted in significantly different outcomes, with this setup being rated similarly to the virtual robot condition. This suggests that partial simulation of study setups in HRI research can yield valid results regardless of the level of virtualization, though perceptual challenges must be considered, particularly for tasks requiring precise perception and interaction.

Previous research indicates that participants are more cautious with real robots due to collision concerns, while virtual robots are associated with lower perceived risk [5, 6, 27, 28]. Our work confirms and extends these findings by highlighting the role of task virtualization. Specifically, we found that the fear of collision is

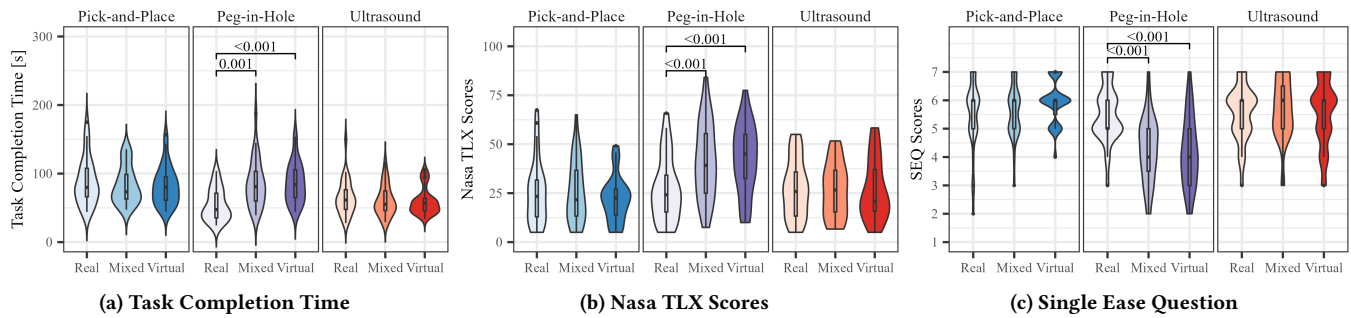


Figure 3: Violin Plots of experimental results of dependent variables.

reduced even when only the task is virtual, while still using a real robot. This highlights the importance of both the robot's and the obstacles' virtuality. However, despite participants' subjective feedback, no significant impact was observed on the recorded measures. In contrast, for the peg-in-hole task, the *real* condition resulted in the best TCT, TLX, and SEQ. This suggests that the slower pace potentially linked to more cautious behavior was overshadowed by other confounding factors. Participant feedback indicated that perception, particularly depth perception, played a crucial role. While perceptual challenges in XR are a relevant topic with ongoing research efforts to address them [1, 13], our findings highlight the task dependence of this limitation. As no significant differences were found between virtualizations for broader tasks, and they were perceived as similar, the employed fidelity seems sufficient for these tasks. However, for precise tasks in virtual settings, special attention should be given to visual fidelity. Our results appear to contradict prior findings that XR enhances efficiency in high-precision tasks [24, 41]. While these findings highlight XR's potential even for precision tasks, they should be interpreted cautiously due to differing interaction techniques in real and XR settings.

One limitation of our study is the use of a single fidelity level, which limits the ability to isolate specific factors influencing transferability between conditions. Although our implementation achieved a promising fidelity level based on the taxonomy by Plümer and Tatzgern [27], the use of the HoloLens constrained visual fidelity. While participants described the limited field of view as sufficient, potential concerns remain regarding shadowing and lighting effects. Previous research suggests that certain performance measures may be independent of visual fidelity [34]. However, future studies should investigate the impact of varying fidelity levels on the transferability of performance between real and virtual tasks. Another limitation stems from the study design itself, focusing solely on absolute validity, assessing whether the real and virtualized use cases yielded equivalent performance. However, related work suggests that even when absolute validity is not achieved, relative order effects may still align [28]. Therefore, future research could investigate whether relative validity holds, particularly for precision tasks where absolute validity is lacking. Additionally it should be noted, that although training sessions were included and task order was counterbalanced, learning effects may have influenced the results. In addition, the participant sample consisted primarily of individuals with an academic background, which may limit

the generalizability of the findings. Future studies should consider including target user groups relevant to the tasks, such as industry professionals or medical practitioners, to identify challenges across different user groups and improve the external validity of the findings.

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